

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First Semester M.Tech. (VDE) (Credit System) Degree Examinations
Supplementary Examination - October - November 2022**21VDE101 – DIGITAL VLSI DESIGN**

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Draw the cross-sectional view of the MOS structure and energy band diagram having p as the substrate for the following modes. (i) Depletion mode (ii) Accumulation mode (iii) Surface inversion mode	10	L*3	1	3
	b)	Calculate V_{OH} , V_{OL} and V_{IL} of the depletion load nMOS inverter having the following parameters. $\mu_n C_{ox} = \mu_p C_{ox} = 25 \mu A/V^2$, $V_{to} = 1.0V$ (enhancement type), $V_{to} = -3.0V$ (depletion type), $\Gamma = 0.4V^{1/2}$, $ 2\Phi_F = 0.6V$, $V_{DD} = 5V$.	10	L3	1	3
2.	a)	Obtain the effect of constant voltage scaling for the following quantities. (i) Capacitor (ii) Drain Current (iii) Power dissipation (iv) Power density. Based on these, comment on the scaling effects on the above parameters in an IC.	10	L2	1	3
	b)	Design a resistive load inverter with $R = 1 k\Omega$ such that $V_{OL} = 0.6V$. The enhancement type nmos driver transistor has the following parameters: $V_{DD} = 5V$, $V_{to} = 1V$, $\Gamma = 0.2 V^{1/2}$, $\lambda = 0$, $\mu_n C_{ox} = 22 \mu A/V^2$. (i) Determine (W/L) (ii) Determine V_{IL} and V_{IH} .	10	L3	1	3
		Unit – II				
3.	a)	Derive the relationship between $(W/L)_n$ to $(W/L)_p$ in case of a symmetric inverter starting from the switching threshold equation of a CMOS inverter.	10	L2	2	3
	b)	Describe the concept of capacitive coupling effect between the interconnect lines.	10	L2	2	3
4.	a)	Obtain the expression for the C_{pHL} of a CMOS inverter.	10	L2	2	3
	b)	Obtain the expression for the average power dissipation in a CMOS inverter.	10	L2	2	3
		Unit – III				
5.	a)	Derive the required capacitance ratio C_{boot}/C_s in case of dynamic bootstrapping circuit.	10	L3	3	3
	b)	Implement 3 input AND gate using Domino CMOS logic and draw the general structure of NORA CMOS logic.	10	L3	3	3
6.	a)	Implement 4 stages ratioed enhancement load dynamic shift register and 4 stages of depletion load nMOS dynamic shift register.	10	L3	3	3
	b)	Construct 4 input NOR gate using BiCMOS logic.	10	L3	3	3
		Unit – IV				
7.	a)	Design 1- bit full adder using transmission gates.	10	L2	4	3
	b)	Show the implementation of 4 bit barrel shifter.	10	L3	4	3

P.T.O.

8. a) Build a comparator from (i) Adder and complementor (ii) Pass logic function.
b) How serial adder circuits are constructed?

10	L2	4	3
10	L2	4	3

Unit – V

9. a) Explain different ESD protection circuits.
b) Draw the diagram which shows the relationship between process and device parameters and the circuit system performance. How the process variation dependent on the drain current of an MOS transistor?
10. a) Show the H tree clock distribution network and 3 level buffered clock distribution network.
b) Define latch-up problem and how the parasitic npn and pnp bipolar transistors are identified in a CMOS inverter circuit.

10	L1	5	3
10	L1	5	3
10	L1	5	3
10	L1	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
